

Linear Functionals and Dual Spaces.

In this section, let $F = \mathbb{R}$ or $\mathbb{C}$ be a field. Let V and W be finite dimensional vector spaces.

Definition: A linear transformation $f: V \rightarrow F$ is called a linear functional on V .

Definition: The set of all linear functionals on V is denoted by $L(V, F)$.

We use $V^* = L(V, F)$, and V^* is referred as the dual space of V .

Proposition: If $\dim V = n < \infty$. Then V^* is isomorphic to $F^{1 \times n}$. The isomorphism can be given by the matrix representation map by fixing a basis of V .

Remarks: If $\dim V = n < \infty$, then $\dim V^* = \dim V = n$.

Examples 1): Let $V = F^n$ and $f: V \rightarrow F$ be a linear functional. Then there exists a unique vector $(a_1, a_2, \dots, a_n) \in F^{1 \times n}$ such that $\forall \vec{x} = \begin{pmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{pmatrix} \in F^n$,
$$f(\vec{x}) = a_1 x_1 + a_2 x_2 + \dots + a_n x_n.$$

Proof idea: Let B be the standard basis. Then $\vec{x} = \begin{pmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{pmatrix}$ has coordinates: $[\vec{x}]_B = \vec{x}$, and $[f(\vec{x})]_B = f(\vec{x})$

On the other hand $f(\vec{x}) = [f(\vec{x})]_B = [f]_B [\vec{x}]_B = [f]_B \begin{pmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{pmatrix}$ with $[f]_B \in F^{1 \times n}$

$\Rightarrow$ Goal: $(a_1, a_2, \dots, a_n) = [f]_B$, i.e. $a_i = f(\vec{e}_i)$ for all $i=1, 2, \dots, n$.

Proof: "Existence." Let $\{\vec{e}_1, \vec{e}_2, \dots, \vec{e}_n\}$ be the standard basis of F^n .

Then take $a_i = f(\vec{e}_i)$ for $i=1, \dots, n$.

Then $\forall \vec{x} = \begin{pmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{pmatrix} \in F^n$, $\vec{x} = \sum_{i=1}^n x_i \vec{e}_i$

$$f(\vec{x}) = f\left(\sum_{i=1}^n x_i \vec{e}_i\right) = \sum_{i=1}^n x_i f(\vec{e}_i) = \sum_{i=1}^n a_i x_i$$

"uniqueness". Suppose $(b_1, b_2, \dots, b_n) \in F^n$ be another vector such that
 $\forall \vec{x} = \begin{pmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{pmatrix}$ such that $f(\vec{x}) = b_1 x_1 + b_2 x_2 + \dots + b_n x_n$.

In particular, for each $i=1, 2, \dots, n$, $f(\vec{e}_i) = b_i \cdot 1 = b_i = a_i$.

$$\Rightarrow (b_1, b_2, \dots, b_n) = (a_1, a_2, \dots, a_n). \quad \square$$

2) . Let $a_1, a_2, \dots, a_n \in F$ be n fixed numbers. Define:

$$f: F^n \rightarrow F$$

$$\begin{pmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{pmatrix} \mapsto a_1 x_1 + a_2 x_2 + \dots + a_n x_n.$$

Then f is a linear functional on F^n .

Check: $\forall c \in F, \forall \vec{x} = \begin{pmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{pmatrix}, \vec{y} = \begin{pmatrix} y_1 \\ y_2 \\ \vdots \\ y_n \end{pmatrix} \in F^n$

$$f(c\vec{x} + \vec{y}) = f\left(\begin{pmatrix} cx_1 + y_1 \\ cx_2 + y_2 \\ \vdots \\ cx_n + y_n \end{pmatrix}\right) = a_1(cx_1 + y_1) + a_2(cx_2 + y_2) + \dots + a_n(cx_n + y_n)$$

$$= ca_1 x_1 + a_1 y_1 + ca_2 x_2 + a_2 y_2 + \dots + ca_n x_n + a_n y_n$$

$$= c(a_1 x_1 + \dots + a_n x_n) + (a_1 y_1 + a_2 y_2 + \dots + a_n y_n)$$

$$= cf(\vec{x}) + f(\vec{y}).$$

3). Let $A \in F^{n \times n}$, define $f: F^{n \times n} \rightarrow F$ by $f(A) = \text{Tr}(A)$

i.e. denote $A = (a_{ij})$, then $f(A) = \text{Tr}(A) = a_{11} + a_{22} + \dots + a_{nn}$.

Then f is a linear functional on $F^{n \times n}$.

Check for any $A = (a_{ij}), B = (b_{ij}) \in F^{n \times n}$ and any $c \in F$.

$$\Rightarrow cA + B = (ca_{ij} + b_{ij}).$$

$$f(cA + B) = \text{Tr}(cA + B) = (ca_{11} + b_{11}) + (ca_{22} + b_{22}) + \dots + (ca_{nn} + b_{nn})$$

$$= c(a_{11} + a_{22} + \dots + a_{nn}) + (b_{11} + b_{22} + \dots + b_{nn})$$

$$= c \operatorname{Tr}(A) + \operatorname{Tr}(B).$$

4). Let $V = \mathcal{P}(\mathbb{R})$ let $t_0 \in \mathbb{R}$ be a fixed number. Define

$$L_{t_0} : V \rightarrow \mathbb{R}$$

$$p(x) \mapsto p(t_0), \text{ i.e. evaluation of } p(x) \text{ at } x=t_0.$$

Then L_{t_0} is a linear function on V .

Check: $\forall p(x), q(x) \in V$ and $\forall c \in \mathbb{R}$. Then $cp(x) + q(x) \in V$.

$$\begin{aligned} \Rightarrow L_{t_0}(cp(x) + q(x)) &= cp(t_0) + q(t_0) \\ &= c L_{t_0}(p(x)) + L_{t_0}(q(x)). \end{aligned}$$

5). (Homework) Let $V = C[a, b]$ be the set of all continuous functions on $[a, b]$

Let $t_0 \in [a, b]$ be a fixed number. Define $L_{t_0} : V \rightarrow \mathbb{R}$ by $L_{t_0}(f(x)) = f(t_0)$

$\forall f(x) \in C[a, b]$. Show that L_{t_0} is a linear functional.

Theorem: Dual basis. Let V be an n -dimensional vector space. Let $B = \{\vec{v}_1, \vec{v}_2, \dots, \vec{v}_n\} \subseteq V$

be an ordered basis. Then there exists a unique dual basis $B^* = \{f_1, \dots, f_n\}$

of V^* such that $f_i(\vec{x}_j) = \delta_{ij}$, i.e.

$$f_i(\vec{x}_j) = \begin{cases} 1, & \text{if } i=j. \\ 0, & \text{if } i \neq j. \end{cases}$$

$$\text{i.e. } f_1(\vec{x}_1)=1, f_1(\vec{x}_2)=0, \dots, f_1(\vec{x}_n)=0$$

$$f_2(\vec{x}_1)=0, f_2(\vec{x}_2)=1, \dots, f_2(\vec{x}_n)=0$$

$$\vdots \quad \quad \quad \vdots \quad \quad \quad \ddots \quad \quad \quad \vdots$$

$$f_n(\vec{x}_1)=0, f_n(\vec{x}_2)=0, \dots, f_n(\vec{x}_n)=1$$

$$\text{Moreover, for each } f \in V^*: f = \sum_{i=1}^n f(\vec{x}_i) f_i$$

$$\text{for each } \vec{x} \in V: \vec{x} = \sum_{i=1}^n f_i(\vec{x}) \vec{x}_i, \forall \vec{x} \in V.$$

Proof: As a linear transformation on V is uniquely determined by its action on

a basis of V , thus the existence and uniqueness of $\{f_1, \dots, f_n\}$ is because of its definition.
 Now, we need to prove that $\{f_1, \dots, f_n\}$ is a basis. Since $\dim V^* = n$, and $\{f_1, \dots, f_n\}$
 has exactly n vectors. Thus we only need to show that $\{f_1, \dots, f_n\}$ is linearly independent.

In fact, Suppose $a_1 f_1 + a_2 f_2 + \dots + a_n f_n = 0$.

$$\text{Then } (a_1 f_1 + \dots + a_n f_n)(\vec{x}_i) = a_i f_i(\vec{x}_i) = a_i = 0 \text{ for all } i=1, \dots, n.$$

then $a_1 = a_2 = \dots = a_n = 0$.

$\forall f \in V^*$. Suppose $f = \sum_{i=1}^n a_i f_i$.

$$\text{Then for each } j=1, \dots, n, f(\vec{x}_j) = \left[\sum_{i=1}^n a_i f_i \right](\vec{x}_j) = \sum_{i=1}^n a_i f_i(\vec{x}_j).$$

$$= a_1 f_1(\vec{x}_j) + a_2 f_2(\vec{x}_j) + \dots + a_n f_n(\vec{x}_j)$$

$$= a_j f_j(\vec{x}_j) = a_j$$

$$\text{i.e. } a_j = f(\vec{x}_j) \text{ for } j=1, \dots, n.$$

$$\Rightarrow f = \sum_{i=1}^n f(\vec{x}_i) f_i.$$

$\forall \vec{x} \in V$, Suppose $\vec{x} = \sum_{i=1}^n c_i \vec{x}_i$

$$\text{Then for each } j=1, 2, \dots, n, f_j(\vec{x}) = f_j\left(\sum_{i=1}^n c_i \vec{x}_i\right)$$

$$= \sum_{i=1}^n c_i f_j(\vec{x}_i)$$

$$= c_1 f_j(\vec{x}_1) + c_2 f_j(\vec{x}_2) + \dots + c_n f_j(\vec{x}_n)$$

$$= c_j f_j(\vec{x}_j)$$

$$= c_j.$$

$$\text{i.e. } c_j = f_j(\vec{x}) \text{ for all } j=1, 2, \dots, n.$$

$$\Rightarrow \vec{x} = \sum_{i=1}^n f_i(\vec{x}) \vec{x}_i.$$

Example: This example is to find a polynomial $P_0(x)$ of degree at most n

such that given $n+1$ distinct constants $t_1, t_2, \dots, t_{n+1}$ and $n+1$ constants

$C_1, C_2, \dots, C_{n+1}$, $P_0(x)$ satisfies:

$$P_0(t_1) = C_1, P_0(t_2) = C_2, \dots, P_0(t_{n+1}) = C_{n+1} \quad (*)$$

Method 1: (Math 341 Method):

Suppose $P_0(x) = a_0 + a_1x + \dots + a_nx^n$ such that $P_0(x)$ satisfies (*).

Then $a_0 + a_1t_1 + \dots + a_nt_1^n = C_1$

$$a_0 + a_1t_2 + \dots + a_nt_2^n = C_2$$

$\vdots$

$$a_0 + a_1t_{n+1} + \dots + a_nt_{n+1}^n = C_{n+1}$$

$$\Leftrightarrow \underbrace{\begin{pmatrix} 1 & t_1 & t_1^2 & \dots & t_1^n \\ 1 & t_2 & t_2^2 & \dots & t_2^n \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 1 & t_{n+1} & t_{n+1}^2 & \dots & t_{n+1}^n \end{pmatrix}}_A \begin{pmatrix} a_0 \\ a_1 \\ \vdots \\ a_n \end{pmatrix} = \underbrace{\begin{pmatrix} C_1 \\ C_2 \\ \vdots \\ C_{n+1} \end{pmatrix}}_C$$

Note that A is a Vandermonde matrix, and $\det A = \prod_{1 \leq i < j \leq n+1} (t_j - t_i)$

As $t_1, \dots, t_{n+1}$ are distinct, $\det A \neq 0$

$\Rightarrow$ The above linear system about $a_0, \dots, a_n$ has a unique solution: $\begin{pmatrix} a_0 \\ a_1 \\ \vdots \\ a_n \end{pmatrix} = A^{-1}C$.

Method 2. (Math 441 method of using linear functionals). For simpler notations, we

and of
Jan 29

will work with $n=2$ case.

Refashioning the problem for $n=2$:

Given t_1, t_2, t_3 distinct constants, and C_1, C_2, C_3 constants, find $P_0(x) \in \mathcal{P}^2(\mathbb{R})$

such that $P_0(t_1) = C_1, P_0(t_2) = C_2, P_0(t_3) = C_3$.

Idea!

Idea of Proof: If $\{L_1, L_2, L_3\} \subseteq (\mathcal{P}^2(\mathbb{R}))^*$ and $\{P_1(x), P_2(x), P_3(x)\} \in \mathcal{P}^2(\mathbb{R})$ are dual bases. Then $P_0(x) = \sum_{i=1}^3 L_i(P_0(x)) P_i(x)$.

Inner Products.

Definition: Let V be a vector space over $\mathbb{C}$. An inner product on V is a complex-valued function of two ordered vectors in V , denoted by $(\cdot, \cdot)$, which satisfies the following conditions.

- (I) $(\vec{x}, \vec{x}) \geq 0$ for all $\vec{x} \in V$ and $(\vec{x}, \vec{x}) = 0$ if and only if $\vec{x} = \vec{0}$.
- (II) $(\vec{x}, \vec{y}) = \overline{(\vec{y}, \vec{x})}$ for all $\vec{x}, \vec{y} \in V$.
- (III) $(c\vec{x}, \vec{y}) = c(\vec{x}, \vec{y})$ for all $\vec{x}, \vec{y} \in V$ and $c \in \mathbb{C}$.
- (IV) $(\vec{x} + \vec{y}, \vec{z}) = (\vec{x}, \vec{z}) + (\vec{y}, \vec{z})$ for all $\vec{x}, \vec{y}$, and $\vec{z} \in V$.

Remarks: $(\vec{x}, c\vec{y}) = \overline{(c\vec{y}, \vec{x})} = \overline{c(\vec{y}, \vec{x})} = \bar{c} \overline{(\vec{y}, \vec{x})} = \bar{c}(\vec{x}, \vec{y})$

$\uparrow$ by II $\uparrow$ by II $\uparrow$ by II

Definition: Let V be a vector space over $\mathbb{R}$. An inner product on V is a **real-valued** function of two ordered vectors in V , denoted by $(\cdot, \cdot)$, which satisfies the following conditions.

- (I) $(\vec{x}, \vec{x}) \geq 0$ for all $\vec{x} \in V$ and $(\vec{x}, \vec{x}) = 0$ if and only if $\vec{x} = \vec{0}$.
- (II) $(\vec{x}, \vec{y}) = (\vec{y}, \vec{x})$ for all $\vec{x}, \vec{y} \in V$.
- (III) $(c\vec{x}, \vec{y}) = c(\vec{x}, \vec{y})$ for all $\vec{x}, \vec{y} \in V$ and $c \in \mathbb{R}$.
- (IV) $(\vec{x} + \vec{y}, \vec{z}) = (\vec{x}, \vec{z}) + (\vec{y}, \vec{z})$ for all $\vec{x}, \vec{y}$, and $\vec{z} \in V$.

Definition. A vector space V over F endowed with a specific inner product is called an inner product space. If $F = \mathbb{C}$, we call V a complex inner product space. If $F = \mathbb{R}$, we call V a real inner product space.

Examples: 1) Let $V = \mathbb{R}^n$. $\vec{x} = \begin{pmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{pmatrix}$, $\vec{y} = \begin{pmatrix} y_1 \\ y_2 \\ \vdots \\ y_n \end{pmatrix}$ define $(\vec{x}, \vec{y}) = \vec{x}^T \vec{y} = (x_1, x_2, \dots, x_n) \begin{pmatrix} y_1 \\ y_2 \\ \vdots \\ y_n \end{pmatrix}$
 $= x_1 y_1 + x_2 y_2 + \dots + x_n y_n.$

2) Let $V = \mathbb{C}^n$. $\vec{x} = \begin{pmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{pmatrix}$, $\vec{y} = \begin{pmatrix} y_1 \\ y_2 \\ \vdots \\ y_n \end{pmatrix}$ define: $(\vec{x}, \vec{y}) = \vec{y}^* \vec{x} = (\bar{y}_1, \bar{y}_2, \dots, \bar{y}_n) \begin{pmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{pmatrix}$
 $= x_1 \bar{y}_1 + x_2 \bar{y}_2 + \dots + x_n \bar{y}_n$

The above inner products on $\mathbb{R}^n$ or $\mathbb{C}^n$ are referred as standard inner product

3) Let $V = \mathbb{R}^{n \times n}$. define $(A, B) = \text{Tr}(A^T B)$.

Check: i). $(A, A) = \text{Tr}(A^T A) \geq 0$ b/c $\text{Tr}(A^T A) = \sum_{j=1}^n \sum_{i=1}^n a_{ij}^2 \geq 0.$

and $(A, A) = 0$ if and only if $A = 0$. (Homework problem result)

II). $(A, B) = \text{Tr}(A^T B) = \text{Tr}(\underbrace{(A^T B)^T}) = \text{Tr}(B^T A) = (B, A)$

b/c Trace of a matrix equals to the trace of its transpose.

III). $(cA, B) = \text{Tr}((cA)^T B) = \text{Tr}(cA^T B) = c \text{Tr}(A^T B) = c(A, B)$

IV) $(A+B, C) = \text{Tr}((A+B)^T C) = \text{Tr}(A^T + B^T)C = \text{Tr}(A^T C + B^T C)$
 $= \text{Tr}(A^T C) + \text{Tr}(B^T C) = (A, C) + (B, C).$

4). Let $V = \mathbb{C}^{n \times n}$. Check that $(A, B) = \text{Tr}(B^* A)$ defines an inner product on $\mathbb{C}^{n \times n}$.

5). Let $V = C^0([-1, 1])$ be the vector space of the set of continuous real-valued functions on $[-1, 1]$. Define

$$(f, g) = \int_{-1}^1 f(x) g(x) dx.$$

Check: i). $(f, f) = \int_{-1}^1 f(x)^2 dx \geq 0$. and $(f, f) = 0$ if and only if $f = 0$.

II) $(f, g) = \int_{-1}^1 f(x) g(x) dx = \int_{-1}^1 g(x) f(x) dx = (g, f)$

III) $(cf, g) = \int_{-1}^1 cf(x) g(x) dx = c \int_{-1}^1 f(x) g(x) dx = c(f, g)$

$$\begin{aligned}
 \text{iv) } (f+g, h) &= \int_1^1 (f(x) + g(x)) h(x) dx = \int_1^1 [f(x)h(x) + g(x)h(x)] dx \\
 &= \int_1^1 f(x)h(x) dx + \int_1^1 g(x)h(x) dx \\
 &= (f, h) + (g, h).
 \end{aligned}$$

Proposition. Let V be an inner product space over $F = \mathbb{C}$ or $\mathbb{R}$. Then for any $\vec{x}, \vec{y}$ and $\vec{z} \in V$ and $c \in F$. The following statements are true.

- 1). $(\vec{x}, \vec{y} + \vec{z}) = (\vec{x}, \vec{y}) + (\vec{x}, \vec{z})$
- 2). $(\vec{x}, c\vec{y}) = c(\vec{x}, \vec{y})$
- 3). $(\vec{x}, \vec{0}) = (\vec{0}, \vec{x}) = 0$
- 4). $(\vec{x}, \vec{x}) = 0$ if and only if $\vec{x} = \vec{0}$
- 5). If $(\vec{x}, \vec{y}) = (\vec{x}, \vec{z})$ for all $\vec{x} \in V$, then $\vec{y} = \vec{z}$.

Proof. Exercise.

Definition. Let V be an inner product space. For $\forall \vec{x} \in V$, we define the norm (or the length) of $\vec{x}$ by $\|\vec{x}\| = \sqrt{(\vec{x}, \vec{x})}$.

Proposition. Let V be an inner product space over F . Then for all $\vec{x}, \vec{y} \in V$ and $c \in F$, the following statements are true.

- 1) $\|c\vec{x}\| = |c| \|\vec{x}\|$
- 2). $\|\vec{x}\| = 0$ if and only if $\vec{x} = \vec{0}$. In any case, $\|\vec{x}\| \geq 0$
- 3). (Cauchy-Schwarz Inequality) $|(\vec{x}, \vec{y})| \leq \|\vec{x}\| \cdot \|\vec{y}\|$
- 4). (Triangular Inequality) $\|\vec{x} + \vec{y}\| \leq \|\vec{x}\| + \|\vec{y}\|$.

Proof: 1). $\|c\vec{x}\|^2 = (c\vec{x}, c\vec{x}) = c\bar{c}(\vec{x}, \vec{x}) = |c|^2 \|\vec{x}\|^2$
 $\Rightarrow \|c\vec{x}\| = |c| \|\vec{x}\|$.

$$3). \|\vec{x}\|^2 = (\vec{x}, \vec{x}) \geq 0 \text{ and } (\vec{x}, \vec{x}) = 0 \text{ iff } \vec{x} = \vec{0}$$

Review on complex numbers: If $z = a + bi$ then $\bar{z} = a - bi$, and $|z|^2 = a^2 + b^2 = z\bar{z}$
 If $z = \frac{z_1}{z_2}$ then $\bar{z} = \frac{\bar{z}_1}{\bar{z}_2}$.

In particular, if $z = (\vec{x}, \vec{y})$. Then $\bar{z} = (\vec{y}, \vec{x})$ then $z \cdot \bar{z} = (\vec{x}, \vec{y})(\vec{y}, \vec{x}) = |\vec{z}|^2$

$$3). \text{ Case 1: If } \vec{y} = \vec{0}, \quad |(\vec{x}, \vec{0})| = |0| = 0 \quad \Rightarrow \quad |(\vec{x}, \vec{y})| = \|\vec{x}\| \cdot \|\vec{y}\|.$$

$$\|\vec{x}\| \cdot \|\vec{0}\| = 0$$

$$\text{Case 2: If } \vec{y} \neq \vec{0}, \quad \forall c \in \mathbb{F}: 0 \leq \|\vec{x} - c\vec{y}\|^2$$

$$= (\vec{x} - c\vec{y}, \vec{x} - c\vec{y})$$

$$= (\vec{x}, \vec{x}) - (\vec{x}, c\vec{y}) - (c\vec{y}, \vec{x}) + (c\vec{y}, c\vec{y})$$

$$\text{Take } c = \frac{(\vec{x}, \vec{y})}{(\vec{y}, \vec{y})} = \frac{(\vec{x}, \vec{y})}{\|\vec{y}\|^2} \quad \Rightarrow \quad = \|\vec{x}\|^2 - \frac{(\vec{y}, \vec{x})}{\|\vec{y}\|^2} (\vec{x}, \vec{y}) - \frac{(\vec{x}, \vec{y})(\vec{y}, \vec{x})}{\|\vec{y}\|^2} + \frac{(\vec{x}, \vec{y})(\vec{y}, \vec{x})}{\|\vec{y}\|^4} \|\vec{y}\|^2$$

$$\Rightarrow \bar{c} = \frac{(\vec{y}, \vec{x})}{\|\vec{y}\|^2} \quad = \|\vec{x}\|^2 - \frac{(\vec{x}, \vec{y})(\vec{y}, \vec{x})}{\|\vec{y}\|^2} = \|\vec{x}\|^2 - \frac{|(\vec{x}, \vec{y})|^2}{\|\vec{y}\|^2}$$

$$\Rightarrow \|\vec{x}\|^2 - \frac{|(\vec{x}, \vec{y})|^2}{\|\vec{y}\|^2} \geq 0 \quad \Rightarrow \quad \|\vec{x}\|^2 \cdot \|\vec{y}\|^2 \geq |(\vec{x}, \vec{y})|^2$$

$$\Leftrightarrow |(\vec{x}, \vec{y})| \leq \|\vec{x}\| \cdot \|\vec{y}\|$$

Review: If $\bar{z} = a + bi$ with $a, b \in \mathbb{R}$. $\text{Re}(z) = a$, $\text{Im}(z) = b$.

$$\text{Then } z + \bar{z} = 2a = 2\text{Re}(z) \leq 2|z|$$

$$z - \bar{z} = 2bi = 2i\text{Im}(z) \leq 2|z|$$

$$\begin{aligned}
 4). \quad \|\vec{x} + \vec{y}\|^2 &= (\vec{x} + \vec{y}, \vec{x} + \vec{y}) = (\vec{x}, \vec{x}) + (\vec{x}, \vec{y}) + (\vec{y}, \vec{x}) + (\vec{y}, \vec{y}) \\
 &= \|\vec{x}\|^2 + 2\operatorname{Re}(\vec{x}, \vec{y}) + \|\vec{y}\|^2 \\
 &\leq \|\vec{x}\|^2 + 2|\langle \vec{x}, \vec{y} \rangle| + \|\vec{y}\|^2 \quad \downarrow \text{by the above result.} \\
 &\stackrel{\text{Cauchy-Schwarz}}{\leq} \|\vec{x}\|^2 + 2\|\vec{x}\| \cdot \|\vec{y}\| + \|\vec{y}\|^2 \\
 &= (\|\vec{x}\| + \|\vec{y}\|)^2
 \end{aligned}$$

$$\Rightarrow \|\vec{x} + \vec{y}\| \leq \|\vec{x}\| + \|\vec{y}\|. \quad \square$$